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Task Description:	This year was a transition year for the systems group researchers (investigators, postdoctoral staff, graduate students, and technical staff) from learning and exploratory mode, which can be mainly characterized by proof-of-concept studies to build a viable toolbox of systems analysis methodologies suitable for ALS analysis, e.g. SimOpt (SIMulation based OPTimization), Markov Theory and Model Predictive Control, towards well structured and coordinated efforts to improve their readiness level for repeated use in answering questions that matters to NASA and ALS community at large. As systems group was undergoing this transition, the need for expanding the scope of systems group, which was set limited only to ALS-NSCORT technologies at the original center proposal, is recognized. Consequently, a supplemental proposal in collaboration with Dr. Drysdale has been developed and submitted to NASA. This proposal was crafted to expand the systems group's analysis and modeling scope to all ALS technologies and to adopt lifecycle management view to study 15-20 year Mars Base ALS system evolution which recognizes the fact that Mars Base ALS system will not stay as it was deployed at the very first mission. Unfortunately, support and resources requested in the supplemental proposal could not be made available at that time as NASA was undergoing through major re-organization. However, another unfortunate development within the systems group, resignation of Dr. Trimble of Howard University from ALS-NSCORT, made some funds available at the ALS-NSCORT to pursue a portion of the activities proposed in the supplemental support request. Consequently, Dr. Lasinski and Dr. Russell joined to the systems group as fulltime Post-Doctoral Research Fellows to fill the void generated per successful completion of Dr. Applequist's Post-Doctoral appointment, and to support the expanded scope of the systems group and the implementation of life-cycle management view. Furthermore, a proof-of-concept study has been initiated in collaboration with Envision Center at Purdue University on visualizing the evolution of Mars Base and its ALS system. It is our pleasure to report YanFu Kuo's graduation from the program (who has successfully fulfilled the MS degree requirements in Mechanical Engineering), and Selen Aydogan's peer reviewed journal publication. Besides its research charter, systems group has also engaged in several outreach activities such as giving invited lectures at the McKenzie Career Center to Mr. Martin's high school class, generating projects based on the ALS problem in Chemical Engineering Design Course (ChE 450) lead by Dr. Pekny, supervising a senior undergraduate student summer research project (Rebecca Alway-Cooper of Chemical Engineering), editing a "baby BVAD" intended to be used in the project "Equivalent Systems Mass Analysis of Plant Growth" with high school students lead by outreach group of ALS-NSCORT. Systems group has also continued leading and supporting center-wide project management, roadmap development, process map revisions and ESM evaluation of ALS-NSCORT technologies initiatives and activities. As systems group we are excited with our new vision and hope that our near-term future work will fuel high impact results for ALS community in addition to scholarly success. Seza Orcun Integrated Systems Group – Focus Area Lead For Systems Modeling of ALS: 1 - Development of a computational framework and analysis for studying the real-life behavior of an ALSS. 2 - Development of a protocol for investigating the possible design configurations for an ALSS. 3 - Demonstrating the performance of the developed computational framework by case studies. For a System dynamics approach to modeling using the Model Predictive Control (MPC) formulation: 1 - ESM estimation for STAR 2 - ESM estimation and comparison for LiFT 3 - Automated pH level control to reduce crew-time requirement for plant growth. For Systems Modeling of an Advanced Life Support System using ESM reduction and safe living environment: 1 - Investigate oxygen, carbon dioxide and water supply, storage tank and treatment capacities required for long term Mars surface mission 2 - Explore possible solutions to reduce the water storage tank ESM through what-if scenarios and sensitivity analysis 3 - Investigate and validate methodology based on Markov Principles, Catastrophe Theory, and possible use of BioSim to ensure a safe working and living environment for crewmembers in an enclosed environment for a given mission while minimizing ESM.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of computer models for control of complex processes
	For Systems Modeling; A Simulation Based Optimization Approach: Development of the computational framework and analysis tool. Figure 1 presents the SIMOPT structure developed to observe the overall behavior of the ALS systems. For a specific ALSS scenario, the simulation is replicated 50 times (test were done with different number of runs and it is observed that using more than 50 runs does not affect the initial results significantly), to generate 50 different timelines. This time series data is processed using time-series data mining techniques to determine the safety zones for re-supplies. The safety zone (safety buffer) for a given variable represents the range of acceptable values for that variable for the specific time interval. For example, the safety buffer for O2 re-supply gives the maximum and minimum amount of O2 that would be required by the system for any given day. Safety buffer values are fed to the ALSS optimization module as constraints. The optimal solution, which minimizes the re-supplies that are difficult to procure or transport in a given ALSS scenario and the amount of untreatable waste, is fed as operating conditions to the ALSS simulation and it is replicated several times to see if the data in the created timelines agree with the predetermined safety zones. If the amounts of re-supplies are within the predetermined safety zones (with a specified confidence level) for the components O2, CO2, H2O and food, SIMOPT is terminated generating the safety zone values. However, if trigger events occur, which means that the data collected from the timelines are not within the predicted safety zone values, then the safety zone determination algorithm is called to scan and search the timeline series data to update the safety zone values, which are then sent to the optimization as new constraints. This loop continues until no more trigger event is observed in the system. Three different life support system scenarios with different technology levels are modeled and the overall behaviors of these systems with their impact on the overall waste generation are investigated using the described SIMOPT framework for a 600 day Mars base mission. These scenarios are: (1) Scenario 1: No waste recovery system, where all supplies are shipped from earth, (2) Scenario 2: Physicochemical (PC) waste recovery system, in which food is supplied from earth while significant portion of air and water are regenerated by PC systems, and (3) Scenario 3: A composite bioregenerative and PC waste recovery, where food is grown on site while regenerating air and water. It is concluded that with the current technology levels, for a 600 day mission to Mars, PC waste recovery technology (Scenario 2) seems to be the most promising one. However, bioregenerative & PC waste recovery system (Scenario 3) might become more efficient by processing human and inedible crop wastes separately, resulting in an increase in the waste water recovery efficiencies. Additionally, different options of solid waste recovery system (e.g. bioreactors) should be considered to lower the oxygen requirement of the solid waste utilization for Scenario 3. Therefore, a more hybrid system of bioregenerative and PC technologies might be

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the best option. This is left as a future study. The detailed results of this study can be found in Aydogan et al. (in press). Development of a Crop Planting Scheduling Module. The current version of the SIMOPT framework for ALSS does not allow for scheduling changes throughout the course of the simulation. For example, for a bioregenerative ALSS scenario, if some or all of the crop area is lost during a simulation, it is not replenished until the next planting schedule. However, a new optimization could be performed to determine a new planting schedule at that state of the system and those results can be used for the rest of the simulation. This would increase the utilization of the resources, which is crucial for the ALSS. Therefore, a planting schedule model has been developed to overcome this problem. In this study, optimum crop planting schedule that would minimize the ESM of a bio-regenerative ALSS is determined using an advanced crop planting scheduling model in conjunction with a diet optimization model. Mixed Integer Linear Programming (MILP) models are developed to determine the best crop planting schedule and optimum diet for the crew-members. Given the activity schedule of the crew members, the diet optimization module constructs a diet cycle of 20-30 days that would meet the necessary nutritional requirements observing a predetermined diet variety. In doing so the diet optimization tries to minimize the overall system ESM. Necessary biomass amounts calculated by this model are fed into the crop planting scheduling model as the demands. Given these demands and growth parameters for these crops, the crop planting scheduling model determines the best planting schedule that will optimize the system behavior, i.e., the one that would minimize ESM. Figure 2 gives the graphical representation of the diet optimization and crop planting scheduling models. The diet optimization and crop planting scheduling modules has been tested for a 600 day Mars mission with a crew of six. The crew is assumed to be formed from three men and three women members. The weights of the crew members are determined by randomly sampling from a triangular distribution with a maximum value of 98.5 kg (95th percentile American male), minimum value of 41.0 kg (5th percentile Japanese female) and a nominal value of 70.0 kg (Hanford, 2004, Table 3.3.1). The height is assumed to be proportional to the weight and lastly, it is assumed that the light weight crew members are females. For each crew member, a daily activity schedule is prepared for 30 day cycles. The time allocations for a nominal crew schedule table (Hanford, 2004, Table 3.3.3) are utilized for the preparation of the crew schedules. Throughout the 30 day cycle, it is assumed that each crew member takes two days off as vacations (after completion of their extravehicular activity (EVA) weekdays). In a 30 day cycle, there are four weeks, each one of which has five week-day schedules and two weekend schedules and two vacation days. Additionally, it is assumed that two crew-members, one man and one woman team, does a four hours of EVA for every weekday throughout a week. The MILP formulation of the diet optimization, for described crew characteristics and their activity schedules, is solved using CPLEX (Ilog Inc., 2001). The objective function is 1225 kg ESM for 30 days. This makes the overall ESM of the optimum system as 24500 kg for a 600 day mission. The diet generated for 30 days is assumed to repeat until the end of the 600 day period. Using ingredients of the each recipe in the diet, the demands for each crop are calculated. The crop planting scheduling formulation, given the demand and the crop data, calculates the best crop planting schedule, i.e., the one that would minimize the overall re-supply. The total amount of re-supply that is required for 600 days, including the supplies that can not be grown on site, is 1480.8 kg. The total amount of supplies that can not be grown on site add up to 581.6 kg and the rest of the re-supply is to cover the first cycle time of each crop. The proof of concept study showed that these models can be used to predict the planting schedule for a bio-regenerative ALSS and the crop planting schedule can be linked to the crew activity schedules.

For Systems Modeling of an Advanced Life Support System Model Predictive Control (MPC):

ESM estimation for STAR A detailed initial ESM estimation for STAR with a Mars surface mission is completed and published in ICES 05. Based on the experimental results from the STAR research group, the paper documented the assumptions and methodologies that were used in estimating the ESM. Initial trade studies with similar solid waste treatment processes were also presented. ESM estimation and comparison for LiFT An initial ESM estimation for a LiFT process in the Mars surface mission is completed based on the preliminary study and experimental results. A trade study for LiFT and two competitive technologies, VCD and VPCAR, is performed with the respect of energy consumption. The comparison shows that LiFT is advantageous from an energy consumption point of view and can be a viable alternative for urine water recovery in ALS systems. Sensitivity analysis of the ESM provided design guidelines for the LiFT research group to further reduce the process ESM by improved process design. Automated pH level control to reduce crew-time requirement for plant growth A network ready multiplexed pH level control system for a hydroponic growth chamber was completed. Two conference papers are submitted for consideration, one is for potential presentation at the Habitation 2006 and the other is being reviewed for the 2006 American Control Conference. Significant reduction in crew time can be achieved, with a network closed-loop pH control system. The reduction in crew time has a direct impact in ESM estimation for plant growth chamber. Future Research Directions: System control and energy analysis Model Predictive Control (MPC) formulation will be used to develop a subsystem level control strategy to maintain adequate life support capacity but minimize energy consumption or maximize subsystem level efficiency. A phenomenological dynamic model for the NSCORT water/air treatment systems will be developed based on existing data and estimations. This model will be used to formulate and design the MPC controller. The result of the investigation will provide capacity and efficiency requirements and the associated control strategy to maintain adequate air and water reclamation for the crew. It will also provide the necessary dynamic response for other systems that are directly connected to the air/water treatment systems. Continue pH level control validation Continue validating the effectiveness of the automated pH level control by comparing biomass production with and without automated pH control and identify the optimal pH time history based on plant growth physiology. ESM sensitivity analysis The ESM results from the integrated water/air treatment system will be examined and used to perform a sensitivity analysis. Sensitivity analysis can identify aspect of the system that will provide most significant improvement.

For Systems Modeling of an Advanced Life Support System; ESM Reduction and Safe Living Environment:

A dynamic simulation model is built to investigate the oxygen, carbon dioxide and water supply, storage tank and treatment capacities required for long term mission. 6 crewmembers and 15 years of Mars surface mission duration are assumed in this study. Several what-if scenarios are explored to evaluate the impacts of shipping all the supplies from Earth without regenerative technologies and crops, utilizing regenerative technologies, and growing crops on oxygen, carbon dioxide and water supply, storage tank and treatment capacities. 7-day safety stocks for contingency and different rates of leakage and technology inefficiency are also explored. Storage tank ESM is computed for every scenario. From this study, it is found that 1% of total loss (resulted from leakage and technology inefficiency) from the entire mission duration is roughly equivalent to 55 days of safety stock. In other words, leakage and technology inefficiency has a higher impact on storage tank ESM values than the safety stock for contingency. The study also found that water demand and its corresponding storage tank ESM value are the highest, compared to oxygen and carbon dioxide. Therefore, the subsequent study focuses on the water subsystem only. Food subsystem is considered in this work. A linear program is built to determine the best crop combination for crewmember consumption based on crewmember energy requirement and their crop preference. The result affects the total crop area and water needed to

	grow the crops. Different scenarios are run to evaluate whether the total water supply, treatment and storage capacity in crewmember-and-crop scenario are the sum of water supply, treatment and storage capacity in crewmember-only and crop-only scenarios. A framework has been built based on Markov Theory as a base for further studies. This framework outlines the interaction between crewmembers and plants in terms of oxygen and carbon dioxide exchange between these entities. Three states are defined: Safe, Transient, Risky. These states are defined based on the level of carbon dioxide and oxygen in the enclosed area under scrutiny. The “Safe” state consists of sufficient CO2 and O2 for plants and humans, respectively; the “Transient” state consists of sufficient O2 for humans, but insufficient CO2 for plants, hence, leading to an insufficient O2 production when the plants eventually die; the “Risky” state can be broken down into two scenarios: 1) Insufficient O2 and insufficient CO2. Hence both humans and plants are both in danger. 2) Sufficient CO2 for plants but insufficient O2 for crewmembers. The immediate lack of O2 poses a great threat to the health of the humans. Possible actions are defined for each state should one or more action(s) be taken as an effort to change the state of the system. These actions are common to all the states unless there is a need to define different sets of actions for different states. They are: 1) Do nothing 2) Release O2 from tank 3) Release CO2 from tank 4) Increase light intensity on plants 5) Decrease light intensity on plants Probabilities of “success” are generated based on logical reasoning for the purpose of testing and simulation. “Success” refers to a change of state either 1) from “Risky” to “Transient”, or 2) from “Transient” to “Safe”. Each action (including do nothing) taken while the system is in the “Safe” state can either bring the system to “Transient”, or remain at “Safe”. Each action taken while the system is in the “Transient” state can bring the system back to “Safe”, remain at “Transient” or to “Risky”. Similarly, each action taken while the system is in the “Risky” state can either bring the system back to “Transient” or remain “Risky”. The success (or failure) for every effort made to cause a state change should be measured. A reward system has hence been defined. The Reward, Rij(k) is outlined as follows: Rij(k) = +1 if action k causes the system to transit from Risky to Transient Rij(k)= +1 if action k causes the system to transit from Transient to Safe Rij(k)= +1 if action k causes the system to transit from Safe to Safe Rij(k)= -1 if action k causes the system to transit from Safe to Transient Rij(k)= -1 if action k causes the system to transit from Transient to Risky Rij(k)= -1 if action k causes the system to transit from Risky to Risky Rij(k)= 0 otherwise (if action k causes the system to transit from Transient to Transient) Future Research Directions • Sensitivity analysis is to be conducted to determine the impact of changing one variable on the water storage capacity • What-if scenarios are to be explored to investigate possible alternatives to reduce the storage capacity and supply needed from Earth. • A more rigorous definition of “states” to capture a comprehensive set of elements that can directly affect the health of the plants, the crewmembers, and hence the system of living entities. Look into the possibility of using Fuzzy Logic for state definition. • Define different “time horizons” for simulation. Different types of actions may be taken depending on the mission length, or even control policies based on sampling rate. • Include ESM calculations in the determination of “actions” to be taken during state changes. • Besides ESM, include in the “cost” the effect of each action taken on the system in the long run. • Further evaluation of BioSim to determine if it’s the appropriate simulation software for this research.

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Presentation	Aydogan, S., S. Orcun, G. Blau, J. F. Pekny, and G.V. Reklaitis "Determining Optimum Planting Schedule Using Diet Optimization and Advanced Crop Scheduling Models" Presented at the International Conference on Environmental Systems, Rome, Italy. Jul-2005
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